# CNNs vs Vision Transformers under Data Scarcity

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## Abstract

Convolutional networks embed strong image-specific inductive biases: locality, weight sharing, and translation equivariance. Vision Transformers do not, and must learn equivalent structure from data alone. We ask whether these biases still confer a measurable advantage when training data is scarce, and how that advantage interacts with data augmentation. We train a small ResNet-style CNN and a parameter-matched ViT-Tiny of roughly $271{,}000$ parameters each on CIFAR-10 from scratch, using an identical optimisation recipe at six training-set fractions from $1\%$ to $100\%$ and three random seeds per condition. The CNN outperforms the ViT by between 7.7 and 10.9 percentage points at every fraction. In our single-seed RandAugment ablation, augmentation benefits the CNN more than the ViT at every fraction and slightly degrades the ViT at $1\%$.

## 1 Introduction

Convolutional Neural Networks (CNNs) have dominated modern computer vision benchmarks since AlexNet (Krizhevsky et al., 2012), building on convolutional models such as LeNet (LeCun et al., 1998). A central reason for their success is that their architecture hard-codes visual priors aligned with the spatial structure of images: locality, weight sharing, translation equivariance, and robustness induced by subsampling. The Vision Transformer (ViT) (Dosovitskiy et al., 2021) challenged this view by showing that models with weaker image-specific inductive biases can match or exceed CNNs when pre-trained at large scale. This raises our research question: when training data is scarce, does architectural inductive bias still confer a measurable advantage over a transformer with weaker image-specific priors?

We train a small ResNet-style CNN and a ViT-Tiny on CIFAR-10 at six fractions ($1\%$, $5\%$, $10\%$, $25\%$, $50\%$, $100\%$), with three random seeds per condition. Both models are matched in parameter count and trained under an identical optimisation protocol, so that performance differences can be attributed primarily to architecture rather than capacity or compute. We additionally run a RandAugment ablation to test whether augmentation compensates for the ViT's weaker built-in priors.

Our findings confirm that the CNN's architectural priors confer a substantial advantage under data scarcity, and that this advantage persists even when the full training set is used. The CNN outperforms the ViT by $8.5$ percentage points at the $1\%$ fraction, and the gap remains $7.7$ points at the full $100\%$

fraction. Counter to one natural prediction of the inductive-bias hypothesis, RandAugment yields a smaller lift for the ViT than for the CNN at every data fraction.

## 2 Background and Related Work

The architectural assumptions embodied by our CNN are classically exemplified by LeCun et al. (1998): locality, weight sharing, translation equivariance, and subsampling-induced robustness. Dosovitskiy et al. (2021) challenged the necessity of these convolutional priors by showing that ViTs can match or surpass CNNs when pre-trained at sufficient scale, but also found that ViTs underperform comparable ResNets when trained on smaller datasets without strong regularisation. Touvron et al. (2021) subsequently showed that the data-hungriness of ViTs can be mitigated through stronger training recipes, including RandAugment, Mixup, CutMix, stochastic depth, repeated augmentation, and CNN-teacher distillation. Our contribution is to test these ideas in a tightly controlled, capacity-matched setting on CIFAR-10, where the effects of architecture, data scale, and augmentation can be cleanly disentangled.

## 3 Method

### 3.1 Problem framing

Our aim is to isolate the role of architectural inductive bias in image classification as a function of the amount of training data available. We compare two networks that differ along the inductive-bias axis but are matched on the other axes we can control. The first is a small ResNet-style CNN that embeds locality, weight sharing, and translation equivariance directly into every layer. The second is a ViT-Tiny variant that does not bake these priors into its layers and must learn the relevant spatial structure from the data.

### 3.2 CNN architecture

We implemented a compact ResNet-style convolutional neural network for CIFAR-10 classification. The model receives $32 \times 32$ RGB images and begins with a $3 \times 3$ convolutional stem that maps the 3 input channels to 30 feature channels, followed by batch normalization and ReLU activation. The main body contains three residual blocks. The first block preserves spatial resolution and channel count, the second increases the channel count from 30 to 60 and downsamples using stride 2, and the third increases the channel count from 60 to 120 and also downsamples using stride 2. Global average pooling is followed by a final linear classifier that outputs 10 class logits. The CNN has $271{,}480$ parameters.

### 3.3 ViT architecture

We use a compact ViT-Tiny architecture designed to match the CNN baseline in parameter count while preserving the standard Transformer-based formulation introduced by Dosovitskiy et al. (2021). A $4 \times 4$ patch embedding layer, implemented as a strided convolution, partitions each $32 \times 32$ CIFAR-10 image into an $8 \times 8$ grid of 64 non-overlapping patches and projects each patch into an 80-dimensional embedding space. A learnable class token is prepended to the patch sequence, and learnable positional embeddings of shape $(65, 80)$ are added before the Transformer encoder. The encoder consists of four pre-normalisation Transformer blocks, each with four-head self-attention and a two-layer GELU MLP with expansion ratio three. The final class-token representation is layer-normalised and passed to a linear classifier. The model has $270{,}010$ trainable parameters, within $1\%$ of the CNN baseline.

## 4 Experiments

**Dataset.** All experiments are conducted on CIFAR-10 (Krizhevsky, 2009), which contains $60{,}000$ colour images of resolution $32 \times 32$ across ten balanced classes, split into $50{,}000$ training and $10{,}000$ test images. We carve a stratified $10\%$ slice off the training set to serve as validation set. From the remaining $45{,}000$ images we subsample stratified training sets at six fractions: $1\%$, $5\%$, $10\%$, $25\%$, $50\%$, and $100\%$. Images are normalised using the standard CIFAR-10 mean and standard deviation.

**Training protocol.** Every combination is trained from scratch using AdamW (Loshchilov and Hutter, 2019) with learning rate $1 \times 10^{-3}$, weight decay $1 \times 10^{-4}$, cosine annealing, mini-batch size 128, cross-entropy loss, and 50 epochs. The same seed controls both data subsampling and parameter initialisation.

**Evaluation protocol.** During training we evaluate validation loss and validation accuracy at the end of every epoch and keep the best checkpoint by validation accuracy. This checkpoint is evaluated once on the held-out test set. For each baseline cell we run three independent seeds and report the mean and standard deviation of test accuracy.

**Capacity and compute control.** The CNN and ViT have 271,480 and 270,010 trainable parameters respectively, a difference of well under 1%. Both architectures use the same optimiser, learning rate, weight decay, schedule, batch size, number of epochs, stratified training indices, validation indices, and test set.

### 4.1 Results

Table 1 reports the mean and standard deviation of top-1 test accuracy across three random seeds for both architectures at each data fraction. Figure 1 visualises the corresponding curves.

Table 1: Baseline test accuracy on CIFAR-10 in percent top-1, reported as mean ± sample standard deviation over three seeds.

| | Data fraction | | | | | |
|---|---|---|---|---|---|---|
| Model | 1% | 5% | 10% | 25% | 50% | 100% |
| CNN | $40.7 \pm 1.0$ | $54.5 \pm 0.3$ | $61.2 \pm 0.3$ | $68.9 \pm 0.3$ | $75.1 \pm 0.5$ | $80.3 \pm 0.2$ |
| ViT | $32.2 \pm 1.2$ | $44.8 \pm 0.7$ | $50.3 \pm 0.8$ | $58.5 \pm 0.3$ | $65.2 \pm 0.4$ | $72.6 \pm 0.3$ |

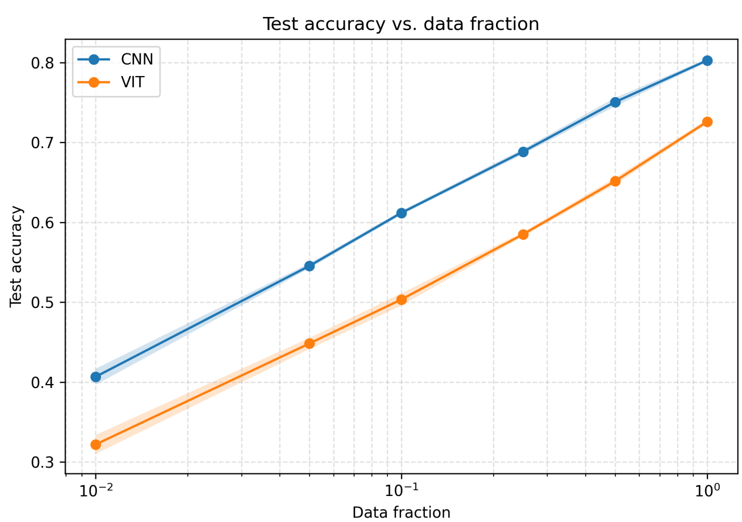


Figure 1: Test accuracy versus data fraction on CIFAR-10. Lines show the mean over three seeds; shaded bands show ±1 sample standard deviation. The horizontal axis uses a logarithmic scale.

Figure 1 shows that both architectures improve monotonically with more training data, rising from 40.7% (CNN) and 32.2% (ViT) at 1% to 80.3% and 72.6% at 100%. The CNN curve lies above the ViT curve at every fraction. The gap ranges from 7.7 to 10.9 percentage points, and the shaded standard-deviation bands do not overlap at any fraction.

### 4.2 Ablations

**Setup.** To isolate the contribution of data augmentation, we re-trained both architectures at all six data fractions with RandAugment (Cubuk et al., 2020) applied to the training pipeline (`num_ops` $= 2$, magnitude $= 9$). All other hyperparameters were held identical to the baseline. Each augmented condition was run with a single seed and compared against the three-seed baseline mean.

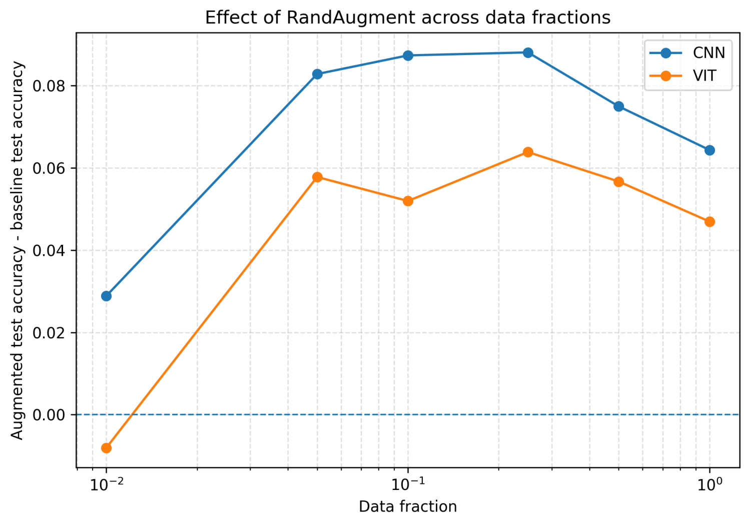


Figure 2: Change in test accuracy when RandAugment is applied during training. Positive values indicate that augmentation improved test accuracy relative to the baseline condition.

**Effect of RandAugment.** Figure 2 shows that RandAugment improves the CNN at every fraction and improves the ViT at five of six fractions, but the ViT's gain is smaller than the CNN's at every fraction. At $1\%$, the ViT's test accuracy decreases by $0.8$ percentage points. Full values are reported in Appendix D.

## 5 Discussion

Our baseline results confirm the first half of the inductive-bias hypothesis: when training data is scarce, the CNN's locality, weight sharing, and translation equivariance are decisive. At the $1\%$ fraction the CNN reaches $40.7\%$ test accuracy compared with the ViT's $32.2\%$. The ViT does not catch up at any fraction we tested. This is consistent with Dosovitskiy et al. (2021), who reported that ViTs only overtake CNNs once pre-training data reaches the JFT-300M regime; with at most $45{,}000$ images, our experimental range lies several orders of magnitude below their crossover.

The ablation weakens the simple substitution hypothesis. RandAugment improves the CNN by between $2.9$ and $8.8$ percentage points at every fraction, while improving the ViT by at most $6.4$ percentage points, and at the $1\%$ fraction it degrades the ViT by $0.8$ percentage points. This suggests that RandAugment alone is not sufficient to compensate for what the ViT lacks. The most natural reconciliation with Touvron et al. (2021) is that the DeiT recipe couples augmentation with other ingredients, including Mixup, CutMix, stochastic depth, repeated augmentation, and a distillation token.

## 6 Conclusion

We asked whether architectural inductive bias still confers a measurable advantage over a transformer with weaker image-specific priors when training data is scarce, and how that advantage interacts with data augmentation. On CIFAR-10, a parameter-matched comparison shows that the CNN's convolutional priors yield a 7.7–$10.9$ percentage-point accuracy advantage over a ViT-Tiny at every data fraction from 1% to 100%, and that RandAugment alone widens rather than narrows this gap. The main limitations are that we evaluate on a single small benchmark, use intentionally small architectures, and run the augmentation ablation with a single seed per condition.

## References


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# A Extended Background and Related Work

The architectural assumptions embodied by our CNN model are classically exemplified by LeCun et al. (1998). In their presentation of LeNet-5, LeCun et al. operationalised a set of convolutional priors that have shaped CNN design ever since: *locality*, since each hidden unit receives input only from a small spatial neighbourhood; *weight sharing*, since units in the same feature map apply the same learned filter across spatial locations; *translation equivariance*, since shifting the input produces a correspondingly shifted feature map; and subsampling-induced robustness to small shifts and distortions. The authors argued that these architectural constraints, combined with end-to-end gradient-based learning, reduce the need for hand-engineered feature extractors by shifting much of feature extraction into a trainable architecture operating directly on pixel images.

Dosovitskiy et al. (2021) challenged the necessity of these convolutional priors by showing that a model with substantially weaker image-specific inductive biases than CNNs can match or surpass state-of-the-art CNNs when pre-trained at sufficient scale. Their Vision Transformer splits an image into fixed-size patches, linearly embeds them, adds positional embeddings, and processes the resulting sequence with a standard Transformer encoder. Unlike CNNs, ViT does not bake locality, two-dimensional neighbourhood structure, or translation equivariance into every layer; these spatial relationships must largely be learned from data. Their experiments established a scaling trade-off: convolutional priors are especially valuable when data are scarce, while their advantage can be offset when enough data are available to learn the relevant visual structure directly.

Touvron et al. (2021) subsequently showed that the data-hungriness of ViTs is not entirely intrinsic. Their Data-efficient image Transformers attain competitive ImageNet accuracy without external pre-training, using ImageNet alone. This improvement comes from a carefully tuned training recipe, including strong augmentation and regularisation techniques such as RandAugment, Mixup, CutMix, stochastic depth, and repeated augmentation. For the strongest models, the recipe also includes a transformer-specific distillation token. In our setting we do not study distillation; instead, our augmentation ablation isolates one component of this broader compensation mechanism.

# B Extended Method Details

## B.1 Problem framing

Both networks are trained from scratch, with no external pre-training, on the same stratified subsets of CIFAR-10, with identical optimisation hyperparameters and almost identical parameter counts. Under these constraints, any systematic difference in test accuracy across data fractions is attributable to inductive bias rather than to capacity, compute, optimisation, or data exposure.

## B.2 CNN architecture

Each CNN residual block consists of two $3 \times 3$ convolutional layers, each followed by batch normalization, with ReLU activations and an additive residual shortcut. When the channel count or spatial resolution changes, the shortcut uses a $1 \times 1$ convolution. This architecture encodes several image-specific inductive biases. First, the use of small $3 \times 3$ convolutional filters encourages locality, because each layer processes nearby pixels and local spatial patterns. Second, convolutional weight sharing means that the same filters are applied across all spatial locations, allowing the model to reuse learned visual features throughout the image. Third, the convolutional structure provides translation equivariance: shifting an image feature tends to shift the corresponding activation rather than requiring the model to relearn the feature at every possible location.

## B.3 ViT architecture

The model operates directly on sequences of image patches rather than on convolutional feature maps. The class token acts as a global aggregation vector whose final representation is used for classification. Because the Transformer architecture itself is permutation-invariant, learnable positional embeddings are added to the sequence to encode spatial information about patch order and location. Dropout with probability 0.1 is applied after positional encoding. Unlike CNNs, Vision Transformers do not hard-code image-specific inductive biases such as locality, weight sharing, or translation equivariance into their layers. Although positional embeddings provide information about patch order, the architecture itself does not inherently assume that nearby pixels are more strongly related than distant ones, nor that visual features should be reusable across locations.

# C Additional Results

Figure 3 shows per-epoch training and validation loss for both architectures over the 50 epochs of training, for the $5\%$ and $100\%$ data fractions. At the $5\%$ fraction, the CNN's training loss decreases to approximately zero within the first 25 epochs while its validation loss plateaus at roughly $1.5$; the ViT's training loss decreases more gradually to approximately $0.3$, and its validation loss reaches a minimum near epoch 10 and then rises again. At the $100\%$ fraction, the train–validation separations are smaller for both architectures.

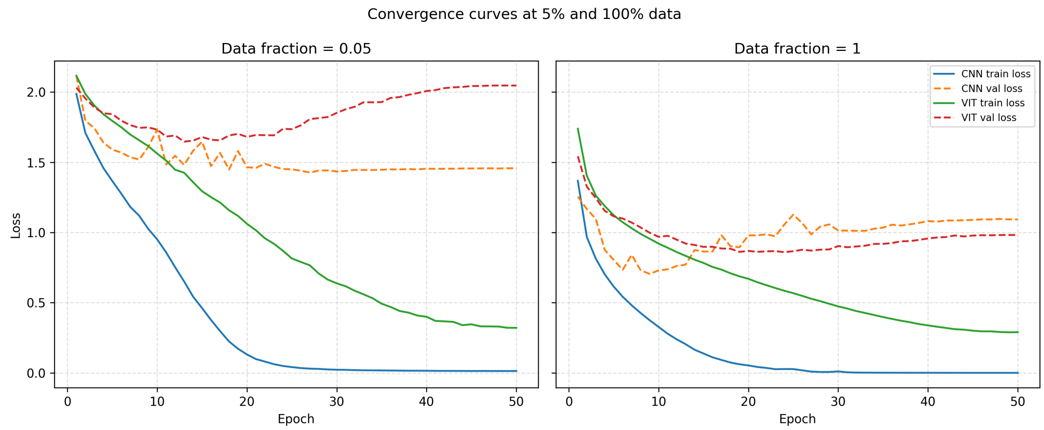


Figure 3: Training and validation loss per epoch at the $5\%$ and $100\%$ data fractions, for the CNN and the ViT.

Figure 4 reports the difference between final-epoch training accuracy and final-epoch validation accuracy, averaged across seeds, as a function of data fraction. For both architectures the gap decreases monotonically as the training fraction grows, from approximately $0.60$ (CNN) and $0.52$ (ViT) at the $1\%$ fraction to $0.20$ and $0.17$ respectively at the $100\%$ fraction.

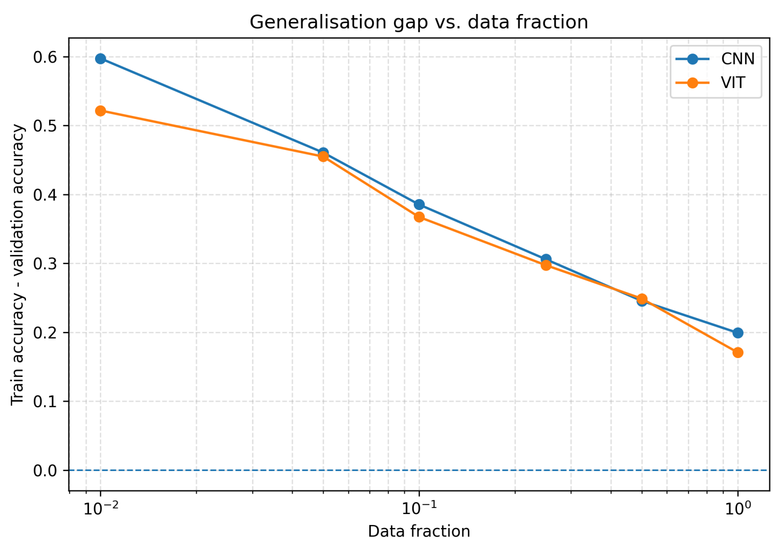


Figure 4: Generalisation gap, computed as final-epoch training accuracy minus final-epoch validation accuracy, averaged across three seeds.

## D Full RandAugment Results

With augmentation enabled, the CNN improves at every fraction: the gain is $+2.9$ percentage points at the $1\%$ fraction, rises to a maximum of $+8.8$ percentage points at the $25\%$ fraction, and remains positive but smaller at the $50\%$ and $100\%$ fractions. The ViT improves at five of the six fractions, by between $+4.7$ and $+6.4$ percentage points; at the $1\%$ fraction, however, the ViT's test accuracy decreases by $0.8$ percentage points.

Table 2: Effect of RandAugment on test accuracy (% top-1). Baseline values are the three-seed mean from Table 1; augmented values are from a single seed. $\Delta$ is the augmented accuracy minus the baseline mean, in percentage points.

| | Data fraction | | | | | |
|---|---|---|---|---|---|---|
| | 1% | 5% | 10% | 25% | 50% | 100% |
| CNN + RandAugment | 43.6 | 62.8 | 69.9 | 77.7 | 82.6 | 86.7 |
| $\Delta$ (CNN) | +2.9 | +8.3 | +8.7 | +8.8 | +7.5 | +6.4 |
| ViT + RandAugment | 31.4 | 50.6 | 55.5 | 64.9 | 70.8 | 77.3 |
| $\Delta$ (ViT) | −0.8 | +5.8 | +5.2 | +6.4 | +5.7 | +4.7 |
| $\Delta_{\mathrm{ViT}} - \Delta_{\mathrm{CNN}}$ | −3.7 | −2.5 | −3.5 | −2.4 | −1.8 | −1.7 |

## E Extended Discussion and Limitations

The gap between training and validation accuracy decreases monotonically for both architectures as more data becomes available. The CNN's generalisation gap is slightly larger than the ViT's at every fraction. This is not evidence that the ViT generalises better in absolute terms: the CNN's larger gap reflects the fact that it drives training accuracy close to 100% on the small training sets, while the ViT cannot fit the training set as tightly.

Our ablation runs return the most surprising finding of the study. The hypothesis that data augmentation partially substitutes for missing architectural priors predicts that RandAugment should help the ViT more than the CNN, particularly in the low-data regime where the priors matter most. We observe the opposite. The augmented gap between CNN and ViT is wider than the baseline gap at every fraction. This weakens the simple substitution hypothesis: RandAugment alone is not sufficient to compensate for what the ViT lacks. They also suggest that at very low data (1%, $\sim 450$ images), the additional variability introduced by RandAugment can be actively harmful for an architecture that lacks the spatial priors needed to anchor it.

Several aspects of our setup limit how widely our conclusions can be generalised. First, we evaluate on a single small benchmark; we cannot rule out that the picture differs at higher resolution or on more complex distributions. Second, both architectures are intentionally small; the inductive-bias trade-off may shift at the scale of ViT-Base or beyond. Third, our augmentation ablation uses a single seed per condition, so we cannot quantify variability for the augmented runs. Fourth, models are trained for only 50 epochs, and the ViT's training loss is still decreasing at this horizon. Finally, we did not implement distillation, so our augmentation experiment does not isolate the question of whether any compensation mechanism can close the CNN–ViT gap at small scale.